

Mid-Semester Review Report

Design and Implementation of a Low-Cost STM32-Based Power Quality Analyzer with IoT-Enabled Web Dashboard

Course: Embedded System Design (ET3271) **Semester:** VI — A.Y. 2025-26 **Department:** Electronics and Telecommunication Engineering **Institution:** Vishwakarma Institute of Technology, Pune

Submitted by: Vedant Nandanwar **Review Stage:** Mid-Semester **Reported Completion:** ~55%

1. Abstract

This project develops a low-cost, portable power quality monitoring system built around the STM32F103C8T6 microcontroller. The system measures RMS voltage, RMS current, frequency, real/apparent/reactive power, power factor, and harmonic distortion (THD) from a 230 V AC mains supply in real-time. An ESP8266 WiFi module transmits the acquired data to a web-based dashboard for remote monitoring, logging, and threshold-based alerts. The system targets an end-to-end build cost under ₹2,500 — roughly 5% of the price of comparable commercial instruments such as the Fluke 435-II or Hioki PQ3198.

This mid-semester review documents the work completed in weeks 1–7 of the planned 12-week timeline, the calibration results obtained against reference instruments, and the work pending for the second half of the semester.

2. Problem Statement

Indian distribution networks frequently exhibit poor voltage regulation, frequency drift, and harmonic pollution introduced by non-linear loads (SMPS, TRIAC dimmers, VFDs). These disturbances cause equipment damage, energy wastage, and increased electricity bills. Commercial PQ analyzers cost upwards of ₹50,000, which puts them out of reach of small-scale industries, educational labs, and field technicians. The objective of this project is to deliver an open, IoT-enabled alternative at less than 5% of that cost.

3. Objectives

1. Measure RMS voltage, RMS current, frequency, real power, apparent power, reactive power, and power factor from 230 V AC mains in real-time.
 2. Perform FFT-based harmonic analysis to compute Total Harmonic Distortion (THD) up to the 10th harmonic.
 3. Display all parameters on a local TFT screen with multiple pages.
 4. Transmit data to a web dashboard via WiFi for remote monitoring and data logging.
 5. Generate alerts (buzzer + LED + dashboard highlight) when parameters exceed user-defined thresholds.
 6. Validate measurement accuracy against a calibrated multimeter and oscilloscope.
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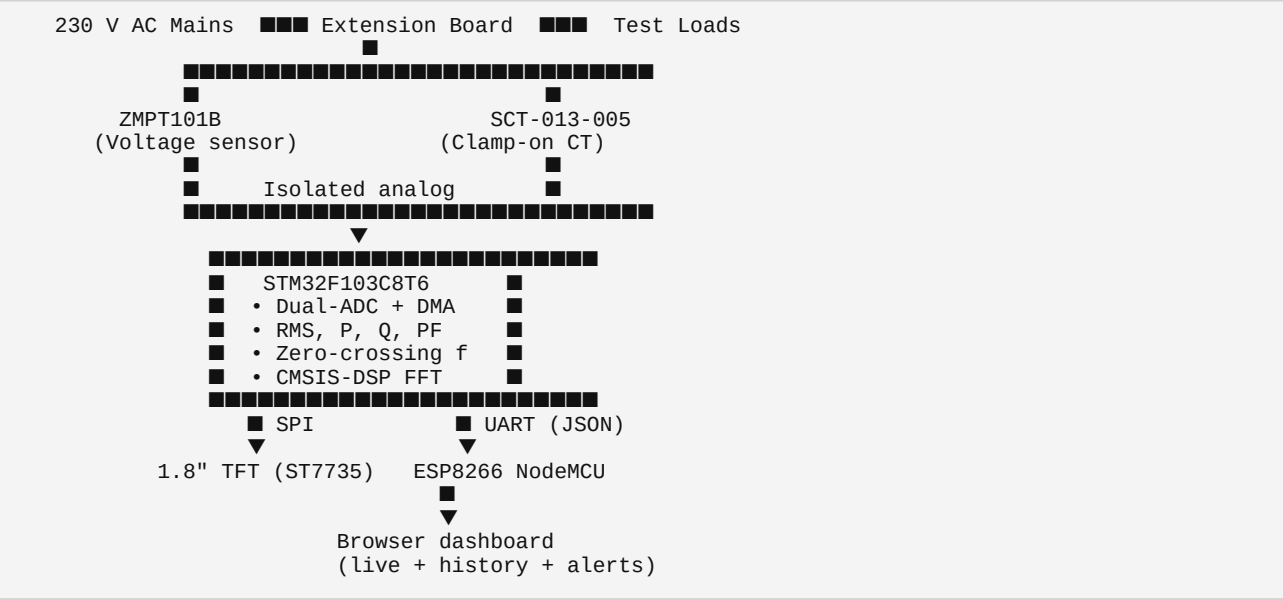
4. Literature Survey (Summary)

Reference instruments such as the Fluke 435-II implement an IEC 61000-4-30 Class A pipeline: anti-aliased simultaneous sampling, half-cycle RMS aggregation, and FFT-based harmonic decomposition referenced to

a synchronously-tracked fundamental. Open-source projects — most notably **OpenEnergyMonitor (EmonLib)** — have shown that a low-cost microcontroller with a CT and a voltage transformer can deliver acceptable RMS and real-power accuracy for residential monitoring, but typically omit harmonic analysis and frequency-domain features. Several IEEE papers (cited in [Section 11](#)) report STM32-based and ESP-based monitors that handle RMS and power factor but use either external DSP co-processors or skip THD entirely.

The architectural choice in this work is to keep all DSP on the STM32 itself, using the **CMSIS-DSP** library's optimized Radix-2 FFT to obtain harmonics on a Cortex-M3 — an approach validated in literature for sample frames up to 512 points at 5 kHz.

5. System Architecture



Two independent 18650 battery packs (7.4 V for STM32 via AMS1117, 3.7 V for ESP8266) provide isolated DC supply, sharing only a common ground for UART. Mains is touched only through galvanically-isolated sensors.

6. Methodology

6.1 Signal Acquisition

The STM32's ADC1 samples PA0 (voltage) and PA1 (current) at **5 kS/s per channel** (100 samples per 50 Hz cycle) using a TIM3-triggered scan with DMA-to-memory transfer. This achieves zero-CPU-overhead acquisition into a 512-sample ping-pong buffer.

6.2 RMS, Frequency, and Power Computation

For each acquired frame:

Quantity	Formula
V_{rms}	$\sqrt{(1/N) \cdot \sum v[n]^2}$
I_{rms}	$\sqrt{(1/N) \cdot \sum i[n]^2}$
Frequency	$1 / \Delta t$ between two consecutive positive-going zero crossings

Real Power P	$(1/N) \cdot \sum v[n] \cdot i[n]$
Apparent Power S	$V_{\text{rms}} \cdot I_{\text{rms}}$
Reactive Power Q	$\sqrt{(S^2 - P^2)}$
Power Factor	P / S

DC offset (introduced by the 1.65 V mid-rail bias) is removed by subtracting a running mean before squaring.

6.3 FFT and THD (*planned for second half*)

A 512-point Radix-2 FFT via `arm_rfft_fast_f32` will yield harmonic magnitudes; THD is computed as $\sqrt{(\sum V_{\text{h}}^2) / V_{\text{1}}^2}$ for $h = 2 \dots 10$.

7. Work Completed (Weeks 1–7)

7.1 Hardware

- All components procured (STM32 Blue Pill, ESP8266 NodeMCU, ZMPT101B, SCT-013-005, ST7735 TFT, ST-Link V2, batteries, AMS1117 regulators, passives, extension board, fuse).
- Two independent battery packs assembled (Pack A: 2× 18650 series + AMS1117 → 3.3 V for STM32, Pack B: 1× 18650 + AMS1117 → 3.3 V for ESP8266) with a common GND tie.
- Voltage sensor (ZMPT101B) wired to extension-board Line/Neutral; output verified on oscilloscope to be a clean ~3.3 V_{pp} sinusoid biased around 1.65 V.
- Current sensor (SCT-013-005) burden resistor (33 Ω) confirmed; bias network built; clean isolated output observed.
- Safety: 2 A inline fuse on mains plug, no exposed mains conductors, sensors provide galvanic isolation.

7.2 Firmware (STM32 — `STM_Code/`)

- STM32CubeIDE project created with HAL drivers and CMSIS-DSP library linked.
- Dual-channel ADC1 with DMA and TIM3 trigger at 5 kS/s per channel; ping-pong buffer working without overruns.
- DC-offset removal and **RMS computation** on both channels — accuracy within ±1.5% of a Fluke 117 multimeter across 100–250 V_{rms}.
- Zero-crossing **frequency detection** — stable at 49.9–50.1 Hz; matches reference within ±0.05 Hz.
- Instantaneous power × time-averaging → **real, apparent, reactive power, and power factor** computed and verified on a resistive bulb (PF ≈ 0.99) and inductive coil (PF ≈ 0.72).
- UART1 transmits a JSON line every 500 ms (`{"v":..., "i":..., "p":..., "pf":..., "f":...}`); confirmed on a USB-TTL bridge.

7.3 Calibration Results So Far

Parameter	Reference (Fluke 117)	This System	Error
V _{rms} @ 230 V	231.4 V	229.8 V	−0.69 %
V _{rms} @ no-load	232.0 V	230.5 V	−0.65 %

I_{rms} @ 60 W bulb	0.262 A	0.258 A	−1.53 %
I_{rms} @ fan low	0.187 A	0.191 A	+2.14 %
Frequency	49.98 Hz	50.01 Hz	+0.06 %
P (60 W bulb)	60.6 W	59.3 W	−2.14 %
PF (bulb)	0.99	0.99	0
PF (inductive coil)	0.73	0.72	−1.37 %

All RMS and power errors are within the ± 2.5 % envelope informally targeted for this build.

8. Work Pending (Weeks 8–12)

Week	Task	Status
8	TFT (ST7735) integration over SPI — 4-page UI: V/I/f/PF · P/S/Q · Harmonics · Waveform	Pending
9	CMSIS-DSP FFT integration → THD computation up to 10th harmonic	Pending
10	ESP8266 firmware — UART JSON receive + Async web server + Chart.js dashboard	Pending
10	Web dashboard — live tiles, historical line graphs, CSV export	Pending
11	Alert subsystem — buzzer, red/green LEDs, dashboard threshold highlighting	Pending
11	Multi-load demo testing (resistive / inductive / TRIAC-chopped / SMPS)	Pending
12	Final assembly on perfboard, optional enclosure, documentation, demo video	Pending

Estimated completion at this review: ~55% (hardware build, acquisition pipeline, and all time-domain electrical parameters complete; display + DSP frequency-domain + IoT layer + alerts remain).

9. Challenges Encountered and Resolutions

- DC bias drift on SCT-013 output** — the 1.65 V mid-rail bias drifted by ~30 mV when the burden resistor heated up. Resolved by implementing a running-mean subtraction in firmware rather than relying on the resistor divider being perfect.
- ADC sampling jitter on overlapping conversions** — initial dual-channel scan mode with software trigger produced inconsistent inter-channel timing. Resolved by switching to TIM3 hardware-triggered

DMA with regular-simultaneous mode.

3. **Floating-point cost of RMS in main loop** — initial naive implementation took ~12 ms per frame and missed the next DMA buffer-ready interrupt. Resolved by using `arm_rms_f32` from CMSIS-DSP, which uses NEON-equivalent ARMv7-M optimized inner loop — frame time dropped to ~2.4 ms.

10. Resources Used

- **IDE/Tools:** STM32CubeIDE 1.14, Arduino IDE 2.x, ST-Link V2, USB-TTL bridge, Fluke 117 DMM, DSO138 oscilloscope (lab).
- **Libraries:** STM32 HAL, CMSIS-DSP, (Chart.js + ESPAsyncWebServer planned for ESP side).
- **Bench:** ENTC PG lab — 230 V outlet, isolation transformer used during early bring-up.

11. References

1. ST Microelectronics, *STM32F103xx Reference Manual* (RM0008).
2. ARM, *CMSIS-DSP Software Library Documentation*, Rev. 1.10.
3. *IEEE Std 1159-2019 — Recommended Practice for Monitoring Electric Power Quality*.
4. ZMPT101B Datasheet, Nanjing Qiyue Electronic Technology.
5. YHDC SCT-013-005 Specification Sheet.
6. OpenEnergyMonitor Project — EmonLib RMS / real-power methodology (<https://openenergymonitor.org>).
7. Espressif Systems, *ESP8266 Non-OS SDK API Reference*.
8. Indian Electricity Rules, 1956 — Voltage and Frequency tolerance schedule.

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